

ActInf MathStream #004.1 ~ David Spivak

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welcome to the active inference institute this is the participatory online laboratory today is tuesday september 13th and we are here with mathstream 4.1 uh with david spivak and we are welcome to take any questions in the youtube chat so if you have any questions please write them there and we will have time to ask our presenter during the course of this discussion uh david is here to talk to us today about dynamic interfaces and arrangements an algebraic framework for interacting systems please take it away okay yeah thanks a lot for having me um so i i'm gonna talk for around 40 minutes but for maybe almost an hour i don't know but feel free to interrupt um with whatever and i wanted to say at the offset that i'm not really talking much about um active inference itself here i'm mainly saying like another perspective on interacting systems or systems that might learn or something like that and i'm more like waiting for the discussion time to kind of find these links so um yeah okay so i want to talk about dynamic interfaces and arrangements and what that means and this applied category theory point of view on things so so why am i here um i'm here to seek a valuable exchange of ideas because active inference offers first time i heard about it it's compelling and most people apparently find it unusual but for some reason it instantly meshed with my own intuition about things that i've gotten from applied category theory and so i think there's a lot we can learn from each other so applied category theory emphasizes structure rather than quantitative analysis so most of math and science today reduce all of our experience to plays of quantity so you have graphs and you have numbers and um you measure things in numbers and you have probabilities and those are numbers and so everything is just a number but experience what's happening now i mean what number would you put on you know the sound of my voice or the or what numbers you could put numbers on this slide but would they tell you what the slide was about and so what category theory emphasizes it's still math but it emphasizes structure it emphasizes relations between things like the relationship between what's on this slide and other things on the slide and which one the rest of the talk the relationship between that and me the relationship between that and you and the kind of coherence between all that like why are we really here what is it um how does it cohere how did it come together so applied

category theory aims to bring that math to relational disciplines outside of math so i want to tell you about applied category theory specifically in reference to interfaces because i'm going to say that we interact through our interfaces when you hear the word interface you might think markov blanket i don't know enough to be sure that that's exactly what i mean so that's kind of where this is fitting in maybe into active inference framing but i'd rather you just think about interfaces um so i'm asking in this talk what sort of algebra could support us in thinking about interaction and stuff like that um i think of mathematical fields as accounting systems so um i like that the word count is in there and i like how kind of humble it is it's like an accounting system here so if you want to um if you have arithmetic that's a mathematical field and it accounts for the flow of quantities like in finance so if you want to your accountant to help you say why you did what you did or what you did and how the quantities flowed you would write something in an arithmetic account you'd write an arithmetic account of it you'd say and they could someone could point that and say what's this 12.83 cents and you could explain so there's numbers and you can account for stuff what if you wanted to account for the teleportation protocol of quantum mechanics or something like that so hilbert spaces is a is a branch of math and it helps you make accounts for the sorts of things that people do in quantum computing or quantum mechanics or generally probability distributions is a mathematical field but it helps you account for likelihoods why is somebody winning in poker and somebody not if you want to give that account and you want to be careful about it you'd use probability distributions so but what is an account of phenomena like we're already kind of talking and you know somebody might give an account of their finances or something and just written in english but you can't really be that careful with it you can't track it you can't be sure you know what they're saying and that you can point at something so the language the accounting system has to articulate the relevant type differences and provide operations that correspond with their interactions and that's really a dense statement um i didn't mean to i'm trying to keep things concise but what do i mean there in in accounting for dollars and like financial stuff there's dollars and there's numbers of things and there and you can add and subtract dollars like this was five dollars and he paid four so there's one dollar left so you can subtract them or add them but you can't multiply two dollar figures it doesn't mean anything to multiply five dollars times eight dollars but you can multiply five dollars by eight things like so i sold eight things of five dollars each so the operations that you're allowed to do um there are different type differences there's the number of widgets and then there's the price of the widgets and that's a difference in type and the operations make sense with plus you're allowed

to add dollars and you're allowed to add numbers of widgets but you're not allowed to multiply uh numbers of dollars by numbers of dollars so that's kind of what i mean here an accounting system needs to tell you like hey that's not what dollars means what would it even mean to multiply five dollars by eight dollars um is it in dollars squared now or something anyway so that's the sort of thing an accounting system does it gives you the um a way of knowing what you're allowed to do what the type differences are and what you can operate how you operate on them so category if every mathematical field is an accounting system then category theory is an accounting system so what is an accounting system for it's an accounting system for the fact that we see in the world coherent structures we see in the world flows of things we see in the world quantum mechanics we see in the world likelihoods and we see in the world coherent structures so if we want to account for them we need some language and the one that's been invented is category theory and so it makes analogies which are similarities of structure with different content i've got one thing and another thing a is to be a c is to be d that similarity is made into a formal object and category theory has been useful in math in computer science physics materials science linguistics um and moreover one thing i didn't see on the slide is that category theory helps you invent new mathematical fields they might not be big ones that are studied by a million people you might just have a micro custom built accounting system written in category theory so you make a new system of structure and what i'm interested in is what sort of system accounting system would could be used to account for dynamic interaction like things interacting so the driving question i really prefer to have a question that i can always be asking instead of one that i need very specific lab to do so i want to know what we have right here i want to account for this incredible world that we have because on earth we have all these amazing forms of life from cells to humans like how did that come about and we have the built world where we have transportation systems and things like that and we have computers it's an amazing built world we have language the ability to systematically present knowledge like a slideshow or a history book or whatever we have morality with rules of thumb for living a good life how did that come about i think in each of these cases they evolved through the push and pull and struggle of living so language is refined by continually using language and the way we present knowledge or the built world the way we architect things um is continually evolving and in even morality like you have a bunch of societies that have to live together and they evolve a way to do that so i want to see see all that um i want to account for that so what are these systems and how they develop and how do we talk about them all at once um what language is appropriate for giving accounts of

this happening in action um and can we use that language to engineer new systems and then something we were talking about just before we started the talk today is like what constitutes health or wisdom what gives the system a sense of direction and how come it keeps finding good senses of direction in the sense that evolution somehow our civilization is offering more today than it did in the past so the only part of this i'm going to discuss is the potential accounting system that i'm offering here or like the beginning of it in other words i don't know how all this stuff happened i just and i don't want to be the one to figure out all the how this stuff happened because obviously it's more than even then you know any group of people could really finish or even and i don't even have the capacity to really start but i think there are many people who do and already are doing that and all i'm trying to offer is a way we could put that into some sort of accounting system okay so this this thing we're doing now is the subject of the talk it's dynamic interaction morphology the way that things form like topos institute formed or a person forms and behavior all that stuff is about little systems that interact and this interaction structure itself changes through time and i'll explain a mathematical language for working with that sort of thing but on some level much of this talk will be a case of its own subject matter in the sense that um i'm interested in the development of the life's order like what is how does this develop but in the science system um this takes place between the interaction between fields so right now uh you guys know much more about active inference and i know more about category theory and so we're participating in this science system where we're interacting and we're having dynamic interaction because after this the part where i'm just talking i'm still looking at blue as she's she's looking at me and i'm seeing her nod and that that helps me decide what to say next and whether we're where um how this communication is working so we're experiencing the development the the ability for systems interacting to shape you know shape that interaction and become sent like sense make together um in real time here so the the concepts discussed in this talk should describe the talk itself as an interaction so let me just pause for feedback and just kind of get someone else's voice into the space if you're willing uh do you see like kind of what i'm saying that the math i'm doing i'm talking about here should be capable i'm not going to give the description because i would be like filling out the spreadsheet with all the facts but it's more like that the map could offer the sort of spreadsheet you would need to talk about this sort of thing so definitely like math related to interaction is really what active inference is all about and fundamentally describes like sense and perception and action um daniel joined us for a minute do you have any extra comments i have a feeling you have something to say here no i would

just ask uh what can math describe and where do art science and other things that are not math also play roles how do those interact great yeah cool so i'm suggesting that math gives you the accounting system and that science fills it in but let me um yeah so i think i'm going to go into that a little more but so there's a self-reflective character where i'm saying that um you guys just actually participated in an interaction with me and like give a kind of sense of direction for this talk a little bit or or at least um one thing we want to discuss which is what's matt what's the role of math and science and so we're kind of creating co-creating uh this sense-making work that we're doing here through our interaction and so that's kind of what i'm saying is that there's a self-reflective thing that the talk itself if you want to you can always dive into this itself as a case um so that's kind of hanging around but i'm not going to push really into that the fact that we're building something together here too much more uh but it's here i i want to talk about how theory compresses and information and then and there's elaboration and there's feedback from experiment um so i think that compression and elaboration play a big role in this story we compress our past into a form that we elaborate in the present so we're constantly forming memories that are not as big as the thing that their memory of but there are compressions that we can elaborate in the present dna for example compresses a record of who died and who thrived in some sense into a language of four nuclear types and um theory any theory is a compression of past experience so the free energy principle is someone has compressed ideas about what they've seen over and over and over again into a very small form and then whether you're talking about dna or memories or the free energy principle these are elaborated in the present so as i speak this theory about compression or whatever not very original but it's here um that is an elaboration of something that compressed into these slides in the past where dna is somehow read by messenger rna and i don't know there's a ribosome involved and stuff and it's elaborated those genes are producing proteins and those proteins are elaborated to are the elaborations that compressed code and they then create the living form of the thing so that seems really important and then similarly experiment in theory if theory is like compression and experiments like elaboration these exist both in biological and mathematical fields so almost everyone knows everyone who knows a biology is knows that there's experiments involved but maybe you wouldn't think that math has experimentation and i think that it does but it's the attempts to articulate and compute things that's how you that's how you are elaborated you talk about things and you compute and we value a formalism f if it makes expression and computation easy so that's the experiment like how good is this thing well does it make it easy to say what i want it doesn't make it easy

to compute so with roman numerals here like i don't think this is very good at expressing but i know it's really bad at computing so if you want to multiply this roman numeral by this roman numeral you're just kind of screwed um there's no procedure you can go through and the only procedure i know is to translate them into hindi arabic numerals 14 and 6 for example and then use what these are built for which is using the distributive law so 14 is 10 plus 4 and um 10 times 6 is easy and 4 times 6 is easy so you get uh 84 and so you translate it back into lxxiv but where that l came from or why there are three x's you'll never figure out it's just 84. so the hindi arabic numerals i'm saying are empirically better like they just win games right like fibonacci not only came up with a one one two three five eight sequence but probably more importantly was the one who brought the hindi arabic numerals so hindus apparently invented them and the arabs saved them for a long time and then fibonacci found them and brought them to to the west and um they're just better they just won they just went out and that's the way that experiment exists in math how easy it is to use and that's the sort of value claim i'm making about what i'm discussing today that it's it's a single accounting system you're not constantly changing what the notation means or um uh changing um how you're thinking about things it's all one system and it just works really well with itself so the plan for the talk is that it'll be in four parts um i'll give a presently available case the thing i'm trying to model i'll give a whirlwind tour of what the math actually looks like i'll talk about some existing applications and open questions and then i'll conclude with the summary okay so the current dynamic arrangement so this zoom call that we're having right now and maybe with the youtube live stream and the people on the chat etc is a dynamic arrangement so how do we think about it i want to break it into three structures interfaces dynamics and interactions so i'm going to have like i think overall six keywords and these are three of them and i'm sorry there are so many but they all are different so each one of us has an interface so it's what i can express and take in i'm expressing through the changes of my vocal cords and the movement of my hands and i'm taking in information through the sensations of my body and my eyes and my ears and then each one of us has dynamics so i have some internal state that you can't see exactly uh but i'm in having in some mood and i'm expecting to read and all that sort of stuff and then that's updated by what i see you guys doing or what i'm reading on the screen and then between us there's interaction which is how my outputs my expression affects you and how your expression affects me so those are what i think is going on a lot of what's going on in this dynamic arrangement between us it's it's really useful to draw boundaries around things so you modularize by nesting reference frames um we have little we reference frames all i

mean by that i don't necessarily mean the thing they mean in relativity theory although i think that would be a case a reference frame is just a box you put around things and you say i can refer to this as alice and this says bob and stuff like that i want a frame where i get to have a notation or a way to refer to things and we have littler systems interacting within bigger systems all the time we could say that uh atom with its you know um electron orbitals or whatever is some kind of uh reference frame or so but atoms are even molecules we could think of the molecule the reference frame for all of its atoms you can point to them and refer to them organizations and societies we modularize and we say that's a thing that's a thing that's a thing and we are putting things together and that's useful in evolution so for example um according to herbert simon like that's how evolution is so fast is because it modularizes our liver and our stomach and let them evolve independently but also or various systems um circulatory or nervous system etc but also uh it's useful in the way we think humans think about things i i want to emphasize again that the map here will not be numerical it'll be structural so as we move into the map you'll see how it manages these reference frames and so like geometry is about shapes geometry doesn't have much number in it but there are numbers a little bit there's like how big the angle is and how big the sides are but it's really about the shapes and the congruences and stuff like that and similarly you can put cat you can put numbers into category theory it accepts that but it's really about the structure of why you're using those numbers and which numbers can be added to other numbers or what what the rules are about the numbers and really you don't need numbers at all the relationships that are happening here for example are not numerical um but before we get there let's make sure we understand our principle subject because we'd like this map to be able to describe things we work with like cells or tadpoles or us or institutes or whatever and soon i'll i'll define what i mean by interface as a mathematical object but i want to ground this talk and something we can all think about together and since we don't have cells or tadpoles here um we're the subject so this this this um zoom call let's say and with the live stream and all that is the subject how do we think about ourselves well each one of us can do certain things and receive certain things okay so i can move and listen we each outwardly express and take in what what we what we do there um with what i'm calling our interface as i said before and i'm going to call these things we the blue will always be the things we can do so it's like the positions of my body and the green will be what i can take in kind of like what's force the forces that act on me or the sensorium i have that acts on me so any expression whether it's a sound or attitude is a kind of position it's a position i'm taking and kind of popular like parlance or whatever

someone's expressing themselves or taking a position but you could also imagine the position of my vocal chords uh moving around and that that movement of position is an expression that comes out as sound your inner states are reflected outwardly as these positions it's happening maybe faster than you can recognize in the case of vocal chords but it's still um an expression using the blue the positions the outward expression and then the world impinges on you or and me and it directs me and it moves me by forces so when i see you know there could be little forces on me or you could just say hey would you mind passing the salt and that impinges on me it directs me to pass the salt it moves me maybe i don't do it but at least it changes my internal state to being being stubborn or something or to being helpful so what your body can receive in a moment this is a kind of weird one i'm not going to emphasize it that much especially later but but it's true i think that what your body can receive in a moment depends on its position so when your eyes are open the type of thing you can get in through your eyes is bigger than when your eyes are closed so more can act on you if i close my eyes i just won't know what you're what you're doing um and when your car goes through a tunnel the gps stops receiving and when you unplug your monitor um from your computer it's no longer or your keyboard is no longer receiving things so the position can affect what it can receive the monitor is no longer receiving from the computer if you disconnect so the story so far is that you output blue positions and you p-l-u-e and you input um forces the force and or sensation or input that you receive changes your internal state and then your state is reflected outwardly as a new position i move my eyes or my head or or my voice box and a new sense so and then if i close my eyes open my eyes i have this whatever happens it may be the same space of sensations or different but i see new sensations and those uh change my internal state etc etc so this is repeat loops repeatedly either discretely if you want to model it that way or continuously but then where do these forces inputs come from uh they come from this interaction so the way that we're arranged here i'm calling it an arrangement that's going to be another keyword zoom arranges it so that my outputs get to you as inputs so you can hear me and if um as you receive what i say your body shifts and that's your output and that comes to me the way this works is based on the arrangement that we're calling zoom and so now we have another color orange and the program arranges it so we can put each other's outputs and our interaction with the program may change the arrangement itself so if daniel wanted to he could spotlight me or mute me or change how the information is passed the very things that you're doing the outputs that daniel is doing or me i could ask him to do that or i could mute myself the part of our outputs have the ability to change the arrangement between us um we

can walk away or whatever you don't have to actually view me i don't know why you're smiling but i'm imagining you doing some uh practical joke or something okay so i'm going to move into the math now um okay so this algebraic theory of interfaces and arrangements there's a bunch of keywords we're going to use and here are the six i think one two three four five six oh seven interfaces which in uh positions forces states arrangements enclosures nesting okay i don't think i've said all of them so far but they shouldn't be too bad at this point so we each have an interface it's what we can it's that through which we interact our interfaces allows ourselves to express ourselves outward to our position and given a position our interface allows us to receive certain forces or sensations our state is changed by the by the received force and then our state is expressed as a position so where state changes based on what we see and then we express it and then multiple interfaces interact together via their current arrangement how we're arranged together it doesn't have to be spatial arrangement we are the three of us here on the zoom call are arranged i guess in space but really more importantly in terms of how zoom wants to arrange us and we also interact with our enclosure which is the thing outside is another interface and these arrangements can be nested hierarchically so here's a picture and it's a kind of schematic it's not supposed to be what we really are but every box here is an interface so i have one two three four five but i also have y one as an interface y two is an interface and a z is an interface and the way way i wanna draw it is that um the interfaces each interface has on the right i'm gonna draw the positions the outputs so this thing is pushing out stuff to the right and on the left it's inputting forces or directions so maybe this is me here and i'm seeing through my eyes and i'm outputting some visual signal and i'm outputting some auditory signal and then somebody else is receiving that auditory signal and seeing some other thing and that other person is saying something and doing something okay so each box will have the positions it can be on the left on the right sorry and the inputs or sensations or forces on the left and my outputs become things that force you or direct you or push you or um are input to you if we're connected like this uh so inside of each box you could imagine there's a state and it's changing based on what's coming in from the left-hand wires so inside of each box is a state and it's it's being affected by what's the left and it's pushing out data on the right and then the multiple interfaces are interacting together in this arrangement so y1 is supposed to be the outer box here this middle-sized red box is y1 the arrangement is this formation of wires here this y1 thing is the enclosure it is enclosing these three things like topos institute encloses some workers inside of it or i enclose um some some organs inside of me or something like that and then these arrangements finally can be nested

hierarchically so there's me and there's you and then we're communicating but inside of us is more stuff and then we're forming this zoom call z okay i hope all that terminology is making sense and feels natural the thing i didn't say here and is very hard to draw is that the arrangements can change through time based on what flows within them so if i say hey please mute me or something like that it might change the arrangement so that you're no longer receiving um from me so the arrangement can change your time based on what flows at the end of this call when we say goodbye it will change the arrangement okay so now we're really moving into the math so as we said an interface consists of two things a set of positions maybe i can be in three positions a b or c or maybe i have 44 real numbers worth of positions i'm just outputting this gigantic 44 dimensional vector constantly or something but whatever you the point is to make an accounting system so you can model what you think uh are being what the positions are maybe you can you can decide for yourself what set of positions uh a being being of type a could have what set of positions being of type b could have so secondly for every position there's a set of forces f_i or inputs you could call the censorium it's what can arrive when you're in that position if you're a protein and you're folded up maybe you just can't receive certain things on your interior you don't receive charge but when you're unfolded you're receiving charge or if you're a hedgehog as uh toby smythe says when you're folded up you can't receive stuff on your belly or something okay so um so what we'll encode this as a polynomial in a single variable y uh why why because it stands for yoneta in category theory and because it doesn't i don't know somehow it's kind of makes you realize this isn't your standard polynomial it's not an x but i don't know i just use y it's traditional for me at this point and this is going to be a polynomial with non-negative coefficients when you think of polynomials you might think about numbers and like x squared of 5 is 25 and stuff like that but i'm not going to be plugging in numbers into these polynomials they're really just ways of housing the data structure or like what we're going to want to work with so i know it's strange but it works really well and it's formal and i don't freak out when you see the sum sign because even though i'm showing you the math i'm not intending for you to like completely follow it or need it it's an accounting system but like fibonacci coming to the west like you might not you know be able to use it right away but but if you're interested you'll you could learn so an interface might be that this would be the whole interface it'd be a set of positions p and this dummy variable y that's always just going to be there because it kind of holds the place for this exponent and the exponent in the position i is the set of forces you can get in position i so now we've made math for this interface thing which has positions and for every position a set

of forces that can depend on the position whether your eyes are closed or open etc so imagine the interface was the sum of y to the fifth y cubed y cubed by cube 62 times y to the zero out of zero you add all those up that would have one position that had five inputs so this sensorium when the person's in this position would be five possible distinctions they could make whereas in these 62 positions they could run around these 62 and be you would watch them running around here and every time you know they are receiving three um possible distinctions as inputs whereas when they're over here they're receiving five when they're over here they're receiving zero they they can't receive ever they can't receive um here's another polynomial r cubed y to the r to the million it satisfies this thing these don't have to be finite sets so in this thing it's positions it runs around r^3 so now it's running around not only on the surface of the earth but it's just running around wherever it wants to in some kind of fixed coordinate system of the world of the universe and at any point no matter where it is it's always receiving exactly a million inputs so maybe it's receiving a thousand by thousand image it's got some so that's what it's receiving but you just make your own it's an accounting system for you to make your own thing your own interface and you can make as many as you want so why do all this craziness with polynomials because what's amazing to me is that the polynomial operations mean things in terms of interfaces we can add polynomials you know how to do that you can multiply polynomials you might remember you do some kind of people call it foil you just kind of multiply the polynomials you can compose two polynomials like p of q but then there's some operations you probably haven't heard of like tensor and or and internal harm and these are just these are what are called monoidal structures on an aminoid closure on a category polynomials and i'm not telling you what that means uh if you know some category theory you can look at it you know already what it means um but uh but they all mean something to us in terms of what you can do to interfaces and i'm just repeating this interface here so if i had one of these called p and you had one of these called q you could add them you could multiply them you could compose them etc so suppose p and q are polynomials representing two different interfaces then what's p plus q well it's another it's another polynomial so it again represents an interface and it would be the sort it would be the interface that can output a position of p or of q and it's sensorium in a given position is whichever if it's a position of p it would be the sensorium of that thing or the force is there and if it's a position of q it'd be the sensorium or forces that you can get there so if p was this interface this kind of thing that output these sorts of positions and input these sorts of forces and q is this one then p plus q is what nine y to the fifth plus two y to the fourth plus y cubed

and that would be like a machine that can be either in one of these three states or one of these three six states and uh sorry um positions and when they are they'd be inputting five possible distinctions uh or one of these two inputting four possible distinctions etc that's what $p + q$ represents it's somehow um making a new interface but $p \times q$ represents making a new interface in a different way something with this interface would output both opposite you have to like remember how you multiply polynomials but basically the sum ends of the product polynomial a sum end of this thing is a sum end of p and a sum end of q so the positions of this thing would be a pair consisting of position of p and a position of q but an input there or a sensation or distinction you can make there would be one of either p or q because like just if you remember like what if p was $5y$ to the fourth outputs five different things and it puts four no matter what and q is $6y^3$ outputting six and inputting three then when you multiply them you would output $30y^7$ i'm just multiplying straight across here $30y^7$ so you output both a position in p and a position in q a position in this five by six grid here and then you'd input one of seven things either something to p or something to q that's just what times does you could say well i want to input to both of them i don't want to input to just one or the other and then you're using a different operation called tensor so $p \otimes q$ takes these two things and multiplies both positions and the directions or and the sensoria um and lets you output both and input both all the time so this operation does that times operation does one i said plus operation does that composition is an interesting one when you compose polynomials what it would do is an output would be something output a p and then also a strategy for no matter what the p input was and output of q it doesn't it doesn't have to wait for you to tell it what the input to p was it'll just give you the strategy for automatically outputting a q thing once you know an input so it runs p then q in series and then yeah and so it uh this thing kind of outputs a protocol but i think i'm going to say that on the next slide a little better so um or pretty soon the or operation runs either p or q or both in parallel and then this bracket operation um runs arrangements for wiring p and q uh if you remember what that means like i'll get into this a little more but this one is going to output arrangements of p in q and input what flows on the wires um so that's going to be an interesting one for us later all i'm really saying is that all these different operations take interfaces and make new interfaces out of them and that allows you to kind of build things up modularly um we're also going to be able to we can also formalize the notion of arrangement we thought about ourselves in the zoom call as an arrangement because zoom lets my outputs be your inputs and vice versa but you could also think about cells and organs and or cells or cell organs um arranged in a cell or cells in a tissue

the arrangement is just how the information passes between those interfaces i'm not so concerned about the spatial arrangement except as it allows information to pass between the interfaces so here's a picture again and the orange thing of the arrangement so two things about this picture you can see that the arrangement is how information passes between interfaces you can kind of see it visually but two things about this will be generalized first the information sharing can be much messier you don't all you don't except in a circuit you don't just have perfect wires the inputs here might be affected by all sorts of stuff and second the information sharing can change like the interfaces can change shape i might close my eyes but also the thing i say to you might change your inner interaction so i'm that's contained in this notion of an arrangement so formally if you do know category theory then you've probably heard of categories functors and natural transformations and polynomials are in fact functors from set to set and an arrangement it looks like this picture but all i mean by it is a natural transformation of polynomial functors so every one of these pictures can be translated into a natural transformation of polynomial functors and uh what this picture would be is you take the little arrangement uh little interfaces p_1 p_2 p_3 p_4 p_5 you would tensor them together and then you the arrangement itself would be a natural transformation from p_1 tensor p_2 tensor p_5 to q the outer outer interface here now what's happening so brief interlude for meta stuff i just told you that an arrangement is natural transformation of and all that's category theory and i'm not expecting you to know category theory i'm not expecting the audience to know category theory so what's up with why why would i even tell you um that stuff i'm just trying to show you what the math looks like let any technical person in the audience who does know category theory see that i'm what i'm talking about and kind of know where to look for what's next but for everyone else or everyone i just want to show what the math looks like and what it's about because so much of math is quantitative or maybe it's about shapes or something but this map is really different so i want you to see it because i want you to ask like what is how is this math and what is it for and the point is it's about interfaces it's about how you manipulate them how you arrange them how you nest them and interfaces have outputs and inputs and they're captured that can depend on the outputs and that sort of structure is kind of hard to talk about cleanly but it's really well captured with polynomials not considered as functions but as like ways of holding structure and then you can do these operations to the polynomials and they get they let you make new interfaces from old and we'll be talking about arrangements and dynamics next but again they'll just be in this category of polynomial functors that um it's all kind of the same story again and again we don't have to change our accounting system to

move between interfaces arrangements nesting dynamics etc okay so dynamics let's look at this arrangement again what if every p_i all these p 's had a dynamical system running it well then what i'm saying here is that q therefore would too so a dynamical system is a thing with states that evolved through time uh my current emotional state or the state of like where i think i am in the talk or whatever is my current state and then i look at you guys and i see what you're looking like or i read the slide and then it changes and then i i output something so that's what a dynamical system is and then it can be formalized as a system of odes or or a discrete dynamical system or you can just think of it as a high level idea and i've said this so many times now but i'll say one more time you're in a state that shows up as your position in other words your output and then any possible input you get would change your state and right so the math could be discrete or continuous it both works in this polynomial setup so mathematically a dynamical system $1p$ is a natural transformation $s \rightarrow p$ where s is the set of states and again why am i telling you this because all the math looks the same whether you're talking about arrangements as natural transformations between polynomial functors or you're talking about dynamical systems themselves inhabiting these interfaces they're all just natural transformations between polynomial functions given the arrangement of all these p 's inside of q which is a natural transformation and giving this now transformation here that puts the dynamics in each of them you get the dynamics in q so $phpi$ had a dynamical system in it so would q by composing natural transformations so this is the currently the last math slide i want to think about dynamics a bit more so if you remember a few slides ago i was talking about operations in a dynamical system on p plus q would be one that can switch between p mode and q mode so sometimes you're outputting p stuff and inputting from it sometimes you're outputting q stuff and inputting from that sorry i'm a little bit sick so my voice is not great today um a dynamical system on p times q would output both simultaneously in any state it's outputting both but we can receive from just one or the other a dynamical system on p tensor q would output both and receive from both a dynamical system on p composed q would be would do the serial protocol where it outputs a p and outputs a strategy for taking any input on p and send and turning it into an output on q and then it would listen for an input on p and an input on q so kind of just tells you here's what i'm going to do no matter what you i mean it gives you a serial protocol is what i'm saying and what about p inner hom q this is the in the category polynomials functors there is a internal harm for tensor you may have absolutely no idea what i'm talking about but it's something that comes with the math that no one created it it wasn't invented it just is part of the structure kind of like the

complex numbers are miraculous in the sense that you um you say you want algebraic closure and all of a sudden you get you get if i have one derivative then i have all derivatives and stuff like that so it's kind of the category of polynomial functors is kind of miraculous it has tons and tons of structure that no one asked it to have it just hasn't but what is this structure this internal harm well it's the sort of thing that runs arrangements for p and q so a dynamical system on this thing where i tensor $p_1 p_2 p_3 p_4 p_5$ comma q and put it in brackets this internal what it does it's a dynamical system this is a polynomial even though it looks like a big expression it's itself a polynomial and this polynomial has outputs and has inputs as usual and so what a dynamical system on this would do is would be output arrangements it output okay now i want you to wire to you and you to wire to you and you do wire to you so if you're forming an institute you'd say wow uh let's have uh this guy report to that person and all that sort of stuff and you would create some arrangement and then that's what you output as the ceo say and then you listen and you see what flows on the wires and you see how alice is talking to bob and how bob is talking to carla and all that stuff and you'd listen to that and as you listen you'd say hey actually i want bob to report to carla instead of alice or whatever and so you would change the arrangement your inner internal state would be to listen and receive the stuff that's flowing on the wires and you'd be affected by that and then you'd output new arrangements for the system and i think that's kind of getting to this morphology thing uh maybe maybe there are systems that are monitoring the the source of transmissions that are happening between subsystems and then uh outputting new ways that they should arrange each other okay so let me give some applications digital circuits and control systems fit neatly here so a computer is a nested arrangement of dynamical systems every transistor the dynamical system and two of them make up a nand gate not end and then you can get an or gate by wiring together three nand gates like this so i've taken my inputs and i split them and i feed them to the nand into the nand and then i put them together and i put them into the nand and now i have or and this whole thing is just like nand has two inputs and one output this whole thing has two inputs and one output but it's got some internal structure and now we've got an or gate we could nest or gates and and gates and nand gates and we can get adder circuits and multiplying circuits and alus arithmetic units whatever and um and cpus and monitors and they're all made just with transistors like the memory cells are all are all like flip-flops or just situated transistors and so we can kind of make this nested system of transistors and get a computer and so the arrangement here is all really simple um it's never changing in time it's soldered in it's fixed unlike us on the zoom call its dynamics are simple every state is

just a function of the input only and it just uses there's no memory in the in the transistors and yet because of loops in the wiring in the arrangement we get um all possible things that the computer can do so with enough nesting and enough um interesting arrangements you even though the arrangements internally are not that difficult when you nest them they get really amazing in the sense you get a computer and that is the dynamical system that is happening right now allowing for this zoom call to take place i guess there's some internet stuff too that i don't know how it's made but the computers are made this way um control systems also have complicated dynamics let's say this is a power plant or something but again i'm i think they basically have fixed arrangement so your cruise control in your car isn't changing whether it's listening to the gas or the um maybe for the battery in a hybrid car there be a changing arrangement but at least old-fashioned control systems have a fixed arrangement of how the feedback's going to work to control the um the speed or whatever temperature um deep learning also fits into the formalism of polynomial functors but in a different way here the interfaces are all the same they're all outputting a real and inputting a real now and the arrangements are all really simple they're activated weighted sums now i i'm drawing this differently than most people would most people would put these three boxes on the left of this big box um and they'd wire the three things in and it would seem like this was a box taking r_3 to r the way i'm thinking about because i want to put this into the context of this accounting system of polynomial functors it works better to think about a neuron listening to some neurons on his left or some predecessor things so like a higher level thing listening to lower level things it makes more sense formalism to put them inside and then the arrangement isn't just a fixed isn't just a wiring diagram but it has more structure as i said those wiring diagrams were just easy cases um often it's more complex but they're still arrangements in the sense of natural transformations of polynomial and what this arrangement would do is it take the output of these sub guys and weight them maybe multiply this one by 3.2 etc add them up and then do an activation function and output that to the higher level bigger thing then when it gets a loss in it would take the gradient and pass the components of that gradient to the three uh or pass the components of that loss to the three things and then take the gradient and change the wiring diagram not the wiring but the interaction pattern so it would change these three numbers so the information flowing out is the inner box's current guess about what's going on in the world the current weighted sum is like the these guys send out their guesses we add them up we send out our guests as the higher level manager here and then we get some loss function we distribute it to our sub units we also use it to change the arrangement um here so

this the changing of weights in the training of a neural network is the uh is the changing arrangements that i've been kind of pushing this whole time um right the arrangement is dynamic the loss not only is sent to little boxes but is updating the arrangement the collection of weights so deep neural networks is one of four examples we currently have of a certain structure that a postdoc working with medium brandon shapiro and i are calling a dynamic organizational structure or dynamic opera um it's a fractal-like system of arrangements that change through time where the arrangement decides how information flows through the system just like i've been saying and yet the flowing information contains the arrangement just like i've been saying and we formalized this using category theory and the four examples we have are deep learning like i was talking about in the previous slide prediction markets um hebbian learning so this is with sophie lipkind and non-cooperative strategic games so we have only four examples of this structure but i'm guessing there are many more so i'm guessing for example that active inference can be another one or maybe free energy minimization or something like that the point is that these are like very fractal structures where like the a collection of deep learners or gradient descenders if you put them together you get another gradient descender if you take a bunch of predictors and make them into a single predictor you get another predictor so like the collection of predictors becomes a predictor and the way that um the way that predictors are the way that wealth or trust is distributed between predictors is properly nested and stuff like that i can't really go into it but uh i'm interested in whether well i'll say that next slide okay so wrapping up kind of many people think of math about number but it's not a math subject an accounting system tracks certain things and what we're trying to track above is dynamic arrangements math is called polynomial functors and it's well known in beloved in category theory and it accounts for interfaces dynamics and arrangements in a really nice way and it's about building a bigger system from little littler parts it accounts for composing circuits and control systems and deep learning and prediction markets and then you could combine these because they're all in one formalism you could take your control system and hit hook it up to a deep learning system and put that in a prediction market and you all of it's just polynomial functions all the way down i think it might have maybe i should have said might have particular value in active inference because it lets us talk coherently and precisely about these things like changing interfaces i i might be wrong but i don't know that that's a part of the formalism of active inference today um in terms of like changing the arrangement and stuff like that but i'm interested to hear more you guys um changing how things communicate change how the communication pattern itself changes as information is exchanged but if we

wanted to make active inference into a dynamic organizational structure for example you might like uh dynamic organizational structure one person when they saw that they thought oh you should really call it multi-scale dynamic organizational structure in order to really clarify your particular brand here um but uh maybe not so especially so what would we do if we wanted to make free energy minimization a dynamic organizational structure well we would specify all the interfaces that we're going to be talking about that would be used to house free energy minimizers like what is a free energy minimizer allowed to output and what are they allowed to input and you would get clear on that as step one and then you would consider how free energy minimizers arrange themselves like what an arrangement of free energy minimizers would be where if you arrange them in that way it would constitute a larger scale free energy minimizer so that's the sort of thing you would have to ask to make this into a dynamic organizational structure and as long as they all cohere nicely there's some formal check we would do we would have a dynamic organizational structure so to summarize i feel like i've said it a million times now but applied category theory is math for tracking interlocking structures cohering structures it's not about how much of something there is it's about how it's arranged and it applies in quantum mechanics and computer science and math and material science and linguistics but it focuses on structural questions in each one it allows the content to be something that someone else handles later and it's just telling you how you're going to be handling that content so that you don't kind of step on each other's toes as you um as you use it so i'm interested i'm interested in the structure of interacting dynamical systems like how i want to account for be able to account for this i don't want to necessarily give the account but i want to be able to account for this stuff that we're doing here together how we change how we're inputting you know i'm changing by turning the speaker off or disconnecting and therefore not inputting the same or how we're outputting you know muting or camera off and how all that affects what happens on this call or wherever we are and how we change and what we learn and how we make sense of things together so again i don't have specifics but what i'm offering is this accounting system this category of polynomial functors to account for dynamic arrangements and i'm hoping that it um is helpful for thinking about active inference so that's it happy to talk about stuff that's super awesome david thank you so much um i have a million questions but i didn't want to interrupt because i was really enjoying just hearing you um because it like helps it gel more solidly in my brain if you can if the interaction is like give me all the things and then let me like pick apart the little pieces and maybe that works for hopefully the audience also uh danielle did you are you watching the

youtube chat is there questions no do you want to go before i go because i'll interrogate him for a while okay and this doesn't block you does it this thing up here no it looks fine okay okay i moved it seems for you to be there lou please begin the interrogation and i'll be ready with the live questions soon cool all right um okay so let me just go back to the beginning um so in the very beginning you said that you feel like active inference kind of meshes with category theory um and i just i'm curious as i mean i think like i kind of got a really good idea through the course of your talk but maybe you could articulate the why um to maybe kind of help us understand where that that basis is coming from on your house cool yeah when i when i first heard it it was one thing was like utility and prediction are melded like what i want is i'm just expecting the world to be a certain way and what i want is for it to be that way so if i want to if i really if i really intend to have something it's because i know that it i kind of deserve it i can't actually i think people might have fantasies where they think they can get things they don't deserve or something but it doesn't work out i'm not sure i feel like i'm going into too much there but but i like this idea of um melding those two things because utility never really made sense to me it's like where do i get this utility function from well but i really what i somehow thought was like like what they're saying is like my my knees are a certain way they're a prediction of like the fact that gravity will be in a certain range or my lungs can take in this sort of or this way because they're a prediction that my lineage has made about what sort of atmosphere i'm going to be put into and um the things i'm saying to you right now like utility whatever i'm using to decide how good it is for me to say a thing is my prediction of what's what's actually wanted here so i like that it like reduced the number of variables there like instead of a number of entities in the universe or something from utility and prediction just to prediction i don't remember like other things exactly but i just remember kind of feeling like oh i also like that it's scale free so free energy minimization for example or or bayesian predict you know updatey stuff doesn't need you to be at a particular scale it's not about subatomic particles or something i like this scale free and that somehow it's a physics-y thing like i'm really looking for a scale-free thing you mentioned emergence blue and i'm going to stop talking soon because i know your lifestyle thing but but um i i'm not a huge fan of the concept of emergence because i feel like it's usually talking assuming we're going to start from a low level and then say and then the next level is magic and then they start from that level they're like and then the next level is magic and i'm really looking for like the principles where the inductive step the move from n to n plus one is the thing we're monitoring so we're never like confused we're like whoa where did this amazing thing happen we're always seeing we're

we're monitoring that part and um i really like that free energy minimization and entropy maximization and stuff like that is the sort of thing that i could imagine happening at all scales no matter if you're a person or you're a cell or you're an atom well definitely i have to apologize for this like beeping what's going on in my background i'll try to remember to mute but i i do think that like yes emergence is cool but also there's this bi-directional information flow um and when we had mike levin come on to the live stream he really talked about it when he discussed his paper the computational boundary of the self so yes like there are bottom up um there are emergent properties that happen in a collective but there's also top down constraints on those emergent properties that come from the quote higher level or maybe like larger scale so like if we're building things up and down there's constraints that go down and information that comes up and you've got to have this bi-directional information flow to even start to think about drawing any kind of boundary or creating like an interface around like the thing itself like that like we we caught we talked about cognitive cones mike love and i don't know if you're familiar with that paper but yeah it was cool cool it's something else and so i know like you've talked with mike a lot and also with chris fields um and when chris came he really talked to us about contextuality and he started to get into some math that i think is maybe related to some of the things that you were talking about here so you talked about operations that correspond to interactions between sets um and chris talked to us about two spaces and like channel theory and as these like interactions that transform sets as um infomorphisms so like uh i i don't know if any of that like language um speaks to to this or or is like it's hard for me to tell what is the language of category theory and and what is like the application of the category theory to something in real life or or that's not just you know the set of x such that x is less than or equal to like these kinds of math words right right yeah so the the two spaces stuff is definitely category theory and relates um there's a nice way of talking about the relationship between um true space and polynomial functors although i don't have it on the tip of my tongue right now so these polynomial functors are kind of because the like if you have these arrangements which are maps of this sort of thing this uh quote unquote higher level thing is talking to the lower level things lower level things are talking to each other and they're also talking to the higher level thing so there's this kind of bi-directional thing there and so on that level i'd say um yes it does sound similar to what mike uh chris fields was saying and then but you also asked like what is the application of category theory um so once you have an accounting system you can build a tool from it so if someone says hey i got this great idea it's all about plus and times and like dollars can be added but not

multiplied and they can be multiplied by numbers but not each other or whatever you make this rule and then someone could build a spreadsheet that program that incorporates that rule maybe current spreadsheets don't allow you don't keep you on the rails with respect to not multiplying two dollar figures but you could imagine that they they did um so what's the application the application is that once you have the accounting system the mathematical accounting system you can build tools from it or you can just think through it on the whiteboard or something but if you build tools uh let's that's a major way people use math um calculus or whatever they simulate things if you use category theory what it lets you do is you say i have this annoyed category called poly and then someone says you know they want to implement it well instead of thinking about all the features they want from an interface language and they're like oh and let's have it print like this and let's have it publish the web like that and like you know they have all these features what it allows you to do is not look at any feature requests you just implement the accounting system you just put it in there just the map and you call it a day and then if somebody else wants to add stuff you do it later and you don't use that to influence kind of the basic code it allows you to write very very highly structured and very compressed code that does that does all of the major back-end work of the thing and then every other feature request is like you know that's kind of the gold plating or like the thing that makes a user for it but um does that make sense as a as an application yeah and like perfectly segues into my next question actually so so you talked about um category theory as like a potential accounting system and early on um even actually i think you even mentioned morality in your talk but even before we were talking about intelligence and like grounding and ethics and wisdom um i'm now like in kernel studying um regenerative finance and potentially like new ways develop like accounting systems that maybe involve like carbon credit or some kind of moral or ethical bound that's like a new system for like not just plus and minus dollars but but maybe like a new um system for accounting that like a ledger that that allows for more thoughtful consideration into um or of like the environment and just the world at large and humanity and our our fate and future as a species um and i just wonder if you've thought about that or um and i definitely think like you definitely think that category theory can maybe jump in and rescue us here or just what are your thoughts on yeah so category theory um you have to do the work of deciding what your accounting system is in some sense like it helps you decide what what things since it has talked about all of math um it can be written in category theory at this point um or at least all that i i can think of it's got all all of math is the stuff that people have used so far to think about the world and so it's somehow our

getting all of that into a single system so it's likely that whatever we're thinking about now with this regenerative finance etc if you know there would be some good way of putting it into category theory but that still has to be done it won't rescue you except it'll be there as like yeah i think i could if i had an expert to work with i could do it um but then in terms of the question itself regenerative finance and um and that and morality and stuff like one i mean i don't know if this is the sort of thing that jumps to mind but one thing i think in our society that might be pretty bad um and it's in ourselves also or at least maybe it's good but it's good in a way i don't understand yet and that is um that in these big arrangements there's little things whose job it is to hide the intentions of the things inside and so um like i think of i'm not sure i'm gonna name this person and it's almost a trigger but um jeffrey epstein i think like what was he good at i kind of think i know he was known for like making these financial derivative instruments that like hid the more the subprime mortgage stuff i i and i feel like his whole island and all the stuff he was doing like he just knew all the secrets and his and i could imagine um backing away from him but like also like the meat industry there are secrets about how meat is produced and when you go to the supermarket their job is to make sure you don't think about that stuff and so um when you form these big arrangements or the financial system like there's a lot of which goes on the financial system that's hiding the almost slavery that takes place in various places um and then in myself maybe the greed i think all this is really coming from the need to survive um at all costs or something like that that seems like our cells in our body don't have but we somehow have gotten um i'm not sure about that so anyway one thing that comes to mind is like what's up with hiding intentions and do is there any place for that wow um also just like from from a very biological perspective i do think that we have little things inside of us maybe because that function it is to hide things but i definitely don't have access to the all of the knowledge of every single cell in my body right or every even organ so so i mean like survive at all costs like yes i could be like knock on some wood growing a little cancer tumor somewhere that's not known to me and like we have mechanisms that mask pain and that even mask like proprio reception or so there's a lot of things going on like there's a lot of hiding even in bile in the biological system that happens between scales which totally leads into my next question so perfectly which was about compression and elaboration um and so so like what is hidden or what is compressed right and and what is the the relationship or function or um utility of category theory in elaborating compression and elaboration like if i get the compressed version of like what the larger box container is what's happening with the larger box can i elucidate the structure and what's happening in the smaller

boxes or the even the arrangement of the smaller boxes in the the within the larger container the larger scale say um and just so so how much how how good is this math is it gonna how how how is it going to help me understand compression and elaboration across scales yeah cool um i first want to say that i think of category theory as humanity's best thought compression language i didn't yeah so for example the category of polynomial functors is full subcategory of functions from set to set spanned by co-products of representables and now i've told you the entire definition and i don't expect you to understand it and i wouldn't even expect of category theories to understand it but i would write that expression that i just said on the board and it would be i don't know 20 words or 30 words that would be sitting there at the top and then i would go through like what do i mean by category what do i mean by set what i mean by functor what do i mean by uh spanned by what do i mean by uh co-product and what do i mean by representable and i would go through those um words and i would explain whichever ones of those you didn't understand and then i would explain those in a compressed way and then you'd say yeah i get that or no i don't get that and we could unpack this thing so the category theory itself before we get to your question about the interfaces and arrangements is is this amazing thought compression language where all the stuff in this talk all the math behind it um and the stuff is all as all an elaboration of of this very very compressed few words um because we really made this language for compressing big thoughts then there if i remember i hope i'm not missing i mean part of the question but you're also talking about the um arrangements themselves as compressions and yeah like an an adder circuit is a compression of a bunch of and or logic circuits that are themselves a compression of nand gates which are themselves transistors arranged in a certain way a certain way certain way and the nand gate if you think about it why did you arrange those two transistors that way because you knew that it would always act this way you can kind of like everyone's ending uh all the time but orr is a pretty good thing to think about and and is a pretty good thing to think about and plus is pretty good and so these compressions out of all the possible ways of arranging transistors into the number of transistors in a outer circuit that can add two three-digit numbers or something like that might i don't know let's just pretend that took um let's just took a thousand transistors out of all the ways of arranging a thousand transistors there's like a few that you would actually like that where you're like instead of adding two numbers and getting like some random thing you get the actual sound and so we compress into for some reason some compressions are useful because we can predict them um and like i can kind of predict your behavior based on what i'm seeing out of on your face and and what you need to know

from me or what i can do to help and vice versa and that ability to compress all this stuff into just something you output on the is is really important and the same is happening in the or gate and the and the outer circuit and stuff so you kind of create things that have a well-defined usefulness even though i have no idea what that means i don't know that does that answer the question or did i miss a major part of it so let me just repeat a part of it so if i am a compression say like let me just do something really simple i am the number six and i'm a compression of you know three numbers like and it could be so can i can i i guess what i compress is the set of the possible numbers that make up six like the three like that that's what i compress right so that's what translates to the compression so you can never really go from the compression number six to figure out which of those three numbers were right right right you can um what you do know is the remaining thing that sickness gives you there's lots of things that use six and they're all happy to know about six and same with or like i don't know how you made your ore you could have made it with 43 000 transistors and wasted a bunch of them but like the point is i don't need to know all that stuff all i need to know is that this thing's going to take to boolean and produce a boolean in a certain way and so the compression this outer interface is saying i don't care about all your private methods just tell me like what can i expect from you and i think part of like you know in our society we have identities i'm guessing that identity which is such a major thing today is like this compression of like what's expected from the person we we are working out like how those identities really do help us expect things from each other so just on the topic of identity and like to follow up and maybe continue this train of thought a little bit so like i am blue i am a human right like i'm a human that's alive that's here talking under the compression of you know math stream number 4.1 which all of this conversation will eventually compress to active institute math stream 4.1 but but like the fact that i am blue and i am human like i am one among the set of many things that is all human and like we describe the set like living human in a certain way like i have a certain number of cells but many of the cells are not human right like so so more cells at some times of your you know digestive cycle than others so more cells sometimes are not human than actually are human and so it's these set of like things like how many human cells how many you know different e coli how many like you know uh different types of bacteria and microorganisms and like all of these things compressed to the set of like living human i would like to be able to unravel like what is like go from living human or like even just find the limits of like okay so now like too many e coli was like some decomposing human that's not alive so so like i would like to be able to really push those boundaries of or

even find the boundary of what makes the set of cells that is a living human versus like when do you it's like the phase transition like the critical point at what critical point are are so many of these microorganisms alive that i become not a living human anymore yeah so in the same way like a an and gate is not made out of and gates it's made out of smaller things or you know a logic all right so we don't really expect the high-level thing to be made out of little things but the reason i think you're saying that there are these things called human cells is because we share this dna and this the dna that you're tracking when you say that hey the e coli is not me and the bacterial gut gut bacteria stuff is not me um because it doesn't have our dna but it is you in the sense that it helps you as a pattern exist and so if i was putting an interface around you i definitely wouldn't personally want to like only use your dna cells and call those you because that would be you wouldn't be a very good you um but uh yeah so what can happen is like these e coli or these cancers or whatever inside of you let's say you're this gigantic arrangement of stuff that what can happen is that they can start to say things that um start to mess with that arrangement and start to change the arrangement so remember the arrangement is changing based on what's flowing through the wires so your e coli things in there might be saying some stuff or spreading some stuff that's changing the arrangement and now this cell is no longer talking about cell when it should have and now that it's no longer talking to that cell because the arrangement has changed now it's now that other cell is not receiving what it usually does so it starts to say different things and slowly this um this changes the arrangement so much that you're no longer a cohering whole and the the things that we can expect from you your identity has changed to one that's like not as interesting definitely and so can you see that in the with the math like where this maybe phase transition takes place or like at what level of rearrangement like maybe it's 20 rearrangement or like you know 20 incoherent arrangement or measure like the information in the circuit transferring from one to like from from one piece to another such that like when the information flow is so disrupted then it like ceases to be a functional arrangement yeah i i think that problem is as hard as like being able to look at a spreadsheet and seeing if there's corruption like people can do it but it's a judgment it's like someone decides like no that's not that's not real or yeah it's like i know it when i see it kind of thing so there might be rules of thumb you could create in this accounting system like uh but i doubt i don't feel comfortable creating like the definition of the faith transition and i wouldn't expect it to be any more than a judgment but cool thank you and thanks for letting me kind of um ramble on a little bit there so on this slide while we're here this is totally changing the subject um if you were roman like if you were ancient

roman using ancient roman numerals would hindi arabic numerals still be empirically better or does that depend on context i think they would i i know that it might not be in the current culture but i think it i think it is just empirically better um like in the sense like is a better baseball player better well they're better at baseball like i'm not saying that's better like maybe you just love these things and it makes you feel really good to see exes and stuff and then for you like it's great like use this thing but if you want to be a math person that multiplies things like you apparently in rome like they were they knew about square root of two back then they figured out like 1.414 apparently i don't know i don't know if that's true but they were pythagoras was like back then like can you imagine taking square roots if this is what you're using it's just you can barely multiply like how could you take e to a power and do could we do um quantum mechanics today if we were having to do this could we do could we do even compound interest with this like i would never ever want like i would so it just wins like it just it's empirically better because those who decide to move to this system will out compete these and um in life there is some sort of competition so i i do think that it's empirically better but i'm happy to hear if you disagree i mean i do think like you know survival of the fittest right so so clearly like the hindi arabic numerals have have won but but also um i think that if i had never seen a hindi arabic numeral before and was trying to multiply that like i would prefer because it's like just what i'm familiar with it's part of my model is to use these roman numerals if i was in ancient rome that would and i would be like what is this one for you know and so it's just whatever i guess that you're you're familiar with but but it's it's um neat to hear your perspective also i mean like if you were a moth and it was turning you know there this the everything was turning brown because of new smoke you might be familiar with being a white moth but like any moth that changed to being a brown moth would win um i don't know yeah there's familiarity and that is a thing because we need that we need homeostasis and stuff but there's i don't know yeah i i think that if someone was a more capable person in terms of being able to change they would appreciate that faculty in themselves like they'd be like ah i'm glad i tried to change yeah well and so now i have a series of questions about um the interface and the dynamics and interactions i don't know if you want to skip forward to those slides so are these are these like terms like you said you had some key terms interface dynamics interaction arrangement um and i'm tracking um these terms are they part of like the formalism of category theory is this something that like everyone is a category they're studying they're not okay so so that was my my question was um you know because the way that you discussed these interfaces positions forces states arrangements enclosures really spoke to active

inference so much which is why like i really get why you're like oh yeah that seems to like kind of mesh with my current ideas um like so like we have the interface which uh we talk about as like a markov blanket like a separation a boundary um and in the markov blanket are like the sense states and the action states so like in the interface it is like you would call it positions and forces like that's where i act on the world and the world acts back on me like in the the markov boundary um and and so similarly like you talked about like the dynamics of a system as being hidden and only like internal to the system and we have like hidden states like your internal states um and then yeah the interaction is like what i would refer to as like maybe a state update like so so there's like you know we interact and then i might change or i might it might just influence my future trajectory so i just wanted to like highlight the great degree of similarity what you talked about and like what we talk about very regularly and active inference great so what i did was i took i i started noticing polynomial functions wanted to do what i was trying to do i took the feeling of using them like when i the intuition i get by asking myself what a map of polynomial functors is or whatever and came up with these words as the things that most trap like most intuitive english words that tracked how the math was was changing or how the math like developed as i made a story about these things so these are really just the words that i found to most give me intuition for tracking what the math was was doing automatically it looks like daniel has his hand raised oh yeah thank you just great questions blue and david thanks for the very powerful and creative talk i think on this topic of the active inference uh ontology mapping i think that was quite well stated which is this idea of an input output system or an interface and things that are on one side of the interface are shielded from that on the other side of the interface otherwise quite literally it would be a different thing it's a really powerful idea and it comes up in computer science with apis it comes up in graphs and topology and we can even think about things like the arithmetic operations like order of operations or nested parentheses it's like well what happens in the parentheses it stays in the parentheses until it's compressed and then it goes into some other area and then also even how you describe searching for intuitive words and i'm sure actively inferring with many learners for terms that have a natural interpretation or a conversational interpretation and also point towards more technical um usage which i think in active inference we see a lot about system delineation and partitioning from the environment like such as the ones that you've brought up nested systems and and also regarding some numerical quantities like surprise attention and so on although also we have recently seen some symbolic active inference implementations from jf klossier that are engaging in this action perception

loop without sending statistical or numerical messages but rather communicating about uh logic values so i just wanted to note that and uh if you have any thoughts otherwise i'll read a question from the chat um yeah a quick thought is like somehow when i we take language for example math and we translate it into motion so like if you say please pass the salt then like 10 to the 15 atoms will move through space you know you say this word in language and then 10 to the 15 atoms move through space and like some salt crystals land on your on your um food or whatever and so by having the math attached into intuitive ideas in our in our minds we can then take the mathematical ideas and translate them into the movement of atoms so like me being able to talk or or whatever and like it's really important that we have intuitive ways of thinking about these things because otherwise they won't translate down into the movement of atoms that actually the action that we see in the world that uh accomplishes anything so i'm i'm i'm worried about all the mathematicians that are doing math that they don't have any sense at all about how it could ever be used and this is kind of the point of topos institute is like you can i don't mind if you do like okay you can do whatever you want but like i don't see a point in doing math if you have no idea how that would ever be useful to anyone even in a thousand years and so i'm i'm i don't know so yeah to connect the point is like to connect with the audience here have this possibly be useful to anyone for real i need to find the words that take the math and translate it into intuitive concepts so that they can come back to the math if they want to or whatever blue anything to add on that go ahead and uh read the chat questions great all right so this will pull us back generalized a little bit and then i'm sure we could talk more active inference or institute talk so ali wrote it is said that category theory is the most abstract area of mathematics out there perhaps we can even call it a kind of meta mathematics daniel dennett claims in intuition pumps and other tools for thinking that we can theoretically go meta on every level of abstraction is it possible to conceive a level of abstraction even more meta than category theory in mathematics yeah great question um there is a category of all categories there is the thing is that when you have a category of all categories first of all it's like really big and second of all um it has an even higher level structure than each category does so it's called a two category of all categories and then there's a three category of all two categories and there but there is amazingly a thing called infinity categories that finishes this thing where there's an infinity category of all infinity categories and i think i think that that the abstract like it's very hard to think about this thing but um and it's still an active area of of research to create like a nice infinity category but the but yeah people keep getting stuck on hey i've got this category i've got all these categories and i want something more abstract in it

and then i needed two category and i ah i'm done i got the two category of all categories but then eventually someone wants the next one and so um so yeah this is this is a great question and there's an active area of research right now okay interesting i'll ask one active inference related question that connects also to enclosures and nesting we've in the generative models that we see in active inference they're bayesian graphs and they're describing control systems and action perception loops and so on we've seen nested models be used in at least two different ways one is to describe systems who are physically nested within one another such as an organism and then the organs the cells the organelles and those interfaces and what that nesting or enclosing can be interpreted as has a natural spatial interpretation however there have also been nested models of cognitive modeling flavors and in those cases it's never been clear whether the nesting or enclosing that's happening is reflecting spatial boundaries in some cognitive space if one chooses that direction of interpretation or in fact what that nesting entails and then importantly why it would have a structural isomorphism with the nesting of the matruska and just things that are physically nested inside of each other so what does nesting and enclosure mean here and is it referring to spatial boundaries or other conceptual boundaries yeah great so um this uh so like same with airplanes or something you have the physical airplane but then different parts of the airplane are really talking to each other that may not be like the the radio of the control tower and the radio of the airplane or in some sense more closely connected than the radio of the airplane and the engine of the airplane the the radio of the airplane and the radio and the pilot are strongly connected and so yeah there's always this logical what i would call physical arrangement and logical arrangement and those might be different but what they are from this point of view or one thing you could do is just so if you if you look at i don't know this x12 and x21 you might imagine that there's a logical way of arranging things where these really seem like they're quite close i probably should have shown this but that picture is the same as this one i've just undone the nesting let's see if i can try to show that yeah let's see this is oh no it's not what oh i've darn it i didn't take the same one that's confusing okay well i think i added an extra loop on this thing or something anyway my point is we can undo this nesting and have it be have this one be closer to this one and some other nesting so you could use this organizational system called an operad to handle the fact that you might want to nest things differently for different purposes this picture here is supposed to be either physical arrangement or logical arrangement it doesn't matter and it's really just about who's talking to who that's the most important thing the thing is that in your liver in your brain in your cell maybe the latency of communication is much better than across

the boundaries sorry for the voice issue well very um interesting and in neuroscience it speaks to the distinctions of structural functional and effective connectivity which if only it was dominoes hitting each other and they were all identical but even in that case i don't think they would be identical so we're dealing with something and that's a great example with the informational and the causal connectivity after all the bayesian causal graph architecture is pointing towards something that is presenting itself as cause with a technical specific definition although a natural word and so it is interesting when the causal architecture may not reflect the physical architecture and vice versa right and this the opera the notion of opera which i never said during the talk this nesting thing is [Music] is designed to allow you to rearrange how you want to consider inside and outside or whether you want to consider physical or logical to to allow you to do that zooming on your own time or in your way in your own way without affecting what the thing is going to do as a whole thank you blue so just to add on to the possible layers of nesting there's also the layer of temporal nesting right so things happen at like a very fast pace that we don't observe like the light is flickering as long as it's like greater than 60 hertz i don't see it flickering but i know it's flickering um and as long as it's like not in my perceptible range then it doesn't really matter to me so so there's um and just similarly like i don't think biological evolution maybe cares all that much about math stream 4.1 but cares about like you know how many children we each have and you know all these other things right so um that that the the relevance the salience across temporal skills is also um something to think about um and we talked about like the changing interface i think you mentioned that um and with with the question of like i don't know if active inference does this changing interface well so it's it does in um the markup kernel which we just actually like learned about and talked about when we were talking about quantum systems with uh chris and mike and um john so that was like i think live stream number 40. also on on that technical note i would suggest that the plain essence of the blanket definition does not entail any sort of dynamical rearrangement however recent work by dalton s and others has pointed towards the definition of a blanket index which reflects a weekly mixing dynamical interface zone that is on a zero to one continuum and that may point towards the kind of rearrangeable interfaces although not with the kind of um category theoretic nature as far as i know but again this is like a very interesting area um and i i can see many applications in systems including online teams and organization and so i really liked that you used what was that hands um an i for making some of these connections clear and just just one last connection uh you had mentioned your applied um applied uh like trails that you're chasing right now and you

mentioned and i do think that um well well first of all you talked about like um before you talked about that you talked about the free energy minimizers right like so if each little component in the system is a free energy minimizer and how like within the fvp it can um scale to form like a container free energy free energy minimizer so yes and that like will stay a free energy free energy minimizer as long as it decreases the free energy of all of the components in the container for it to stay a container right so that like speaks to self-assembly and perhaps morphogenesis and then in these uh so yeah consider how free energy minimizers arrange themselves there we go and how they constitute a larger scale free energy minimizer if it lowers the free energy um and then just in terms of application uh prediction markets i think is a really good way to maybe align category theory with the fep and i think some of that has been done i want to say like the multi-armed bandit problem is maybe like like a rudimentary example of that and i don't remember which live stream that was it was a long time ago but um we did have a a paper dealing with the multi-armed bandit problem and using active inference to solve it and that might be like a way to start to align more on the um just on the uh prediction market area and then david really curious to hear what you think i see a lot of parallels with what you described about transistors again you said a thousand pieces components in this transistor the space of the 1000 pic n connections might be vast and so that's where like selective processes and abductive logic come into play with this two-stroke engine of like diversifying and exploring the space and pruning down based upon function and so we could imagine prediction markets where the nest mate level the the the colony participant is making great predictions but then they're being compiled in a way that's actually inadequate we could also imagine the other extreme where no nest mate knows how much food the colony has or what the weather is outside etc its natural actions still participate in a colony or in a market that makes adaptive decisions so in that structuring of multi-scale sense making the composability and the ability to create systems that are valid across scales and don't enforce any kind of implausible constraint it becomes essential because people point sometimes to wisdom of crowds and all these other related areas and there's probably some areas and some ways of doing it that are just incredibly effective and i'm sure we can think of crowds and ways of doing it and questions where it is just not that way yeah so in this in this work of on dynamic organizational structures there's a kind of fractal-like nature to it where the same thing like there's one formula for the whole thing and so in order to be one of these things like in this technical sense that we wrote up um it can't have all sorts of crazy stuff at one level that doesn't exist at another level it would like we this is not supposed to be an

indictment of it but we tried to make a more socialist example of a prediction market um where the wealth the the the updated wealth would be um more evenly uh distributed and it wasn't hierarchy like it didn't um it didn't have the fractal-like property where it what would happen at the high level was coherent with what was happening to low level and so it could have been because we weren't um we weren't creative enough or whatever but my main point is that there are there are rules to this system that say when it's going to kind of align across the entire structure and when it does then you get a very short description that that happens at all scales what what this kind of tracking system allows is to to say i mean you don't have to trust that dynamic organizational structures are a good system but if you were like well we need some something to guide us here then um you could use it to say well this way of assembling predictions to make a higher level prediction just doesn't work as one of these dynamic organizational structures and therefore we're going to deny it as the way we're going to do stuff because they're just not going to work when we do multi-scale these four examples do so in heavy learning for example what we what we mean by that is um spike time dependent plasticity where like what happens if if i hear you say the same kind of valence as as me but you said it like one second ago whereas i'm saying it now then then i want to wire to you more strongly so this kind of like just you know this kind of time horizon where i'm listening for your you know your agreement with me and if you did before me then i want to wired you more that itself there is a way of putting those together that has this kind of fractal-like structure but there's tons of ways of doing it that don't and so that that's kind of what's going on here in in what you just said with a short description across scales i see it reflected in that definition you wrote up on the chalkboard that could be uncompressed and in past the salt or past the salt police and then in the organizational setting that kind of multi-scale decision making and ability to address a kind of integrated stack of scales such that there is a compression i am running which is describing a coherent set of actions ranging all the way down to the to the cellular and molecular that can be semantically compressed in a verbal or linguistic community in a certain way and there's other people who you say i'm running and they're going to look at you like what language are you speaking so it still is contextual yet we can have these multi-scale descriptions and then i have a note from kevin in the chat in case it hasn't been noted might be worth mentioning to david that markov blankets inactive inference are just a kind of bayesian network directed acyclic graphs with probabilities assigned to the vertices for all those who will be following this trail and working to um build this relationship hopefully between category theory and active inference thank you kevin cool thanks so very

cool i think we're coming up on time uh david do you want to tell us maybe a little bit about or if you have any questions for us about active inference we would be happy to maybe attempt to elaborate um or also if you want to tell us a little bit about topos institute and um what your goal and function or how you came to you know self-assemble in the in that form that would be great well sure yeah um basically in in academia there's no i my co-founder brendan and i wrote a textbook there's no interest in textbook writing in academia i especially like to help people learn a new subject there's no reward for it there's no reward in math for making tools out of the math or like trying to help people either learn it or use it and so um basically i'd been at mit for eight years and then my department had said uh we now have a new policy that you can only be here five years if you don't have tenure and i didn't have tenure i was a research scientist and so he gave me two and a half years to go i was like well okay that makes sense because academia doesn't care about you know um about the stuff that i'm really trying to do and so brendan and i made topos institute because we actually want the stuff we're doing to help the world make more sense of things like we are this gigantic we you know we we're a network of people that the world that has to steer that has to decide is climate change real you know is is obama a muslim we don't know these answers apparently we don't we can't figure these things out and that just i think it's because we don't have the communication to to um to do that and so i just want like i think there's just it's not because i think there is this greed thing in everyone or this need to survive or whatever it is this hiding of information that like is probably causing problems and i don't really understand it and you're right like we don't need everything out in the open but mostly i think what's really causing problems is just lack of communication it's like my greed if i understood how it's affecting you maybe i wouldn't need to do this particular one and like if you understood how the scientists came to the idea of climate change and you could like track track that to the level you wanted to you could like actually you know understand these things then maybe you wouldn't question it it's just that like you haven't done the experiment for yourself and you're not trusting that it's done the right way and so i was hoping at least that the more we can help people communicate and understand each other um and have high fidelity accounting systems so we can like be accountable the more we can solve problems together and like since that wasn't being rewarded in academia itself um we we needed to go outside and position ourselves kind of between academia and industry and society and so that's how we decided to form topos that that notion of accountable communication is really powerful because we see cases where communications are accounted for with meta data logging and peer review and where are we

accountable socially where is it a digital accountability and the accounting schemes are askew and so people are looking at their shred of the spreadsheet uncertain on how those numbers were derived whether it's the whole picture do you have all the sheets in that file like with a partial picture and yet the space of communication strategies just like that thousand part transistor board was vast well there's more than a thousand people and so the space of communication simply edges let alone quality and tone and content is so vast again it emphasizes the need for a vocabulary and a grammar and a syntax of composable dynamic organizational structures so we can even begin to be working not just in this one very local potentially reason in organizational space yeah i totally agree i mean in computers and the internet we have very composable things that you can write really make math out of and like at every scale in some sense there's the same sort of thing happening and this allows computer scientists to talk to each other and the guy working someone working on like internet protocols and someone working on transistors on some level they can kind of talk but like an urban planning person and a you know a biologist geneticist and a mathematician and like i don't know whatever like these people across society they have they're like talking in windings at each other like they're just like you know mailbox envelope ampersand and like they don't understand what each other is talking about and that they're we're never gonna if we can't cohere like there's all these little languages and they're not coherent together and in category theory there's tons of languages there's a language of topology there's a language of algebra there's language of logic there's logic you know there's number theory there's tons and tons of languages and each one needs to be an isolated like needs to have its own space so it can develop notation for itself so it can develop techniques for itself and not have to like ask anybody else's permission and what category theory and same with like urban planning and genetics like they don't the geneticist doesn't want to say like oh sorry i used the word you know structure and i know you're already using the word structure so we'll call it something else like they don't want to like have to do that and yet in category theory what it allows you to do in math is take topology and make an analogy with number theory or with algebra and actually send information that can land intact because you the functor tells you what you're expecting to keep and what you're expecting to lose across this connection and and anything you're expecting keep it keeps and so what we're hoping for is that by creating like finding languages that that you can use across lots of disciplines from say epidemiology to chemistry or whatever um we can create single like languages and then connect them to other languages database languages and physics languages and stuff like that and um allow them to each

evolve independently and create their own independent you know notation and techniques and yet be able to have that interface that compressed thing send something across the boundary to somebody else and have it arrive intact in a trustworthy and accountable way awesome well david thanks for hanging out with us and talking to us about category theory um and maybe come back and talk to us some more about it and maybe we can align deeper on on the relationship between um the type of things that you're discussing and active inference that'd be great that's great yeah thanks a lot for the invitations to be with you guys today great thanks thank you david goodbye bye you